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Universal Test-Time Adaptation for Speech Enhancement

Universelle Test-Time Adaptation für Sprachverbesserung

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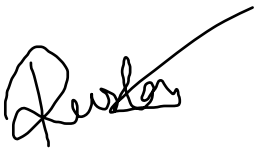
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Stuttgart, August 31, 2026 Rustom Firdosh Bhesania



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Abstract

Speech enhancement is used to recover clean speech from a distorted signal. It is needed wherever speech is captured in real, imperfect conditions, for hearing aids, voice calls, transcription systems, or any system that depends on clean audio as input. The distortions involved are not all the same. Background noise added to a recording is one kind of distortion. Reverberation, caused by sound reflecting around a room before reaching the microphone, is another kind, and behaves very differently from noise.

The recording conditions in the training dataset and the noise conditions at deployment may differ. This is what we call domain shift, and this is why we need models that are domain invariant. Achieving this in practice is difficult, and most speech enhancement models are not trained in this manner. Test-time adaptation (TTA) is where we can adapt these models during the inference phase of deployment. The model updates only a small subset of its weights, without retraining and without labels from the new condition. This is how TTA solves the domain shift problem.

Existing TTA methods for speech enhancement have so far been developed and tested almost exclusively for noise-related domain shift, adapting a model to a new speaker, a new recording setup, or a new language, rather than to a genuinely different kind of distortion. Whether these methods still work when the distortion itself changes, from background noise to reverberation, has not been studied before this research work. To study this, four CMGAN models are trained under controlled conditions: one on noise-related data alone, one on reverberation data alone, and two jointly trained on a custom dataset that interleaves both distortion types in equal measure, differing only in which noise source is paired with the shared reverberation data. Four TTA methods are evaluated across both distortion types on all four models, including a reverb-specific method developed for this research work. None of the three existing methods improve performance when applied directly to reverberation; all degrade it, a result traced to an assumption each method's adaptation signal relies on that holds for background noise but not for reverberation. Jointly-trained models respond far more strongly to adaptation than models trained on a single distortion type alone, traced to adaptation updates concentrating in a shared component of the model whose usefulness depends on joint training having occurred first. Most centrally, adapting a jointly-trained model on one distortion type and evaluating it directly on the other, with no adaptation on the second type at all, produces substantially larger gains than adapting directly on the target type.

These findings suggest that for speech enhancement models facing multiple types of distortion, the most effective adaptation strategy is not always the most direct one.

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List of Abbreviations

ACE	Acoustic Characterisation of Environments (RIR corpus)
CBAK	Composite Background (evaluation metric)
CD	Cepstral Distance (evaluation metric)
CMGAN	Conformer-based Metric Generative Adversarial Network
COVL	Composite Overall (evaluation metric)
CSIG	Composite Signal (evaluation metric)
DEMAND	Diverse Environments Multi-channel Acoustic Noise Database
DIET	Domain Invariant Embedding Transformation
DNS	Deep Noise Suppression (challenge/corpus)
EARS	Expressive Anechoic Recordings of Speech (corpus)
FWSEG	Frequency-Weighted Segmental SNR (evaluation metric)
HPC	High-Performance Computing
IMS	Institute of Natural Language Processing
ISS	Institute for Signal Processing and System Theory
LaDen	Latent Denoising (TTA method)
LLR	Log-Likelihood Ratio (evaluation metric)
MPol	Mask Polarization (TTA method)
OOD	Out-of-Distribution
PESQ	Perceptual Evaluation of Speech Quality (evaluation metric)
RIR	Room Impulse Response
RIRmixIT	RIR-based bootstrapped-remixing TTA method (this project)
SDR	Signal-to-Distortion Ratio
SE	Speech Enhancement
SI-SDR	Scale-Invariant Signal-to-Distortion Ratio (evaluation metric)
SLURM	Simple Linux Utility for Resource Management (job scheduler)
SNR	Signal-to-Noise Ratio
SRMR	Speech-to-Reverberation Modulation Energy Ratio (evaluation metric)
SSNR	Segmental Signal-to-Noise Ratio (evaluation metric)
STFT	Short-Time Fourier Transform
STOI	Short-Time Objective Intelligibility (evaluation metric)
TSCB	Two-Stage Conformer Block
TTA	Test-Time Adaptation
UDA	Unsupervised Domain Adaptation
WHAM!	WHAM! noise corpus

1 Introduction

1.1 Motivation

Speech enhancement is used to recover clean speech from a distorted signal. It is needed wherever speech is captured in real, imperfect conditions, for hearing aids, voice calls, transcription systems, or any system that depends on clean audio as input. The distortions involved are not all the same. Background noise added to a recording is one kind of distortion. Reverberation, caused by sound reflecting around a room before reaching the microphone, is another kind, and behaves very differently from noise.

Deep learning based speech enhancement models, including the CMGAN architecture used throughout this project [1, 2], now perform strongly on this task when evaluated on conditions similar to their training distribution [3, 4].

The recording conditions in the training dataset and the conditions at deployment may differ. This is what we call domain shift, and this is why we need models that are domain invariant. Training a model to be domain invariant directly would require training data that already covers every condition the model might encounter at deployment, which is difficult in practice since the range of possible rooms, microphones, and noise sources a model may face after deployment cannot be fully anticipated in advance, and most speech enhancement models are trained on a fixed, finite set of conditions rather than this full range. Test-time adaptation (TTA) is where we can adapt these models during the inference phase of deployment [5]. The model updates only a small subset of its weights, without retraining and without labels from the new condition. This is how TTA addresses the domain shift problem.

Existing TTA methods for speech enhancement have so far been developed and tested almost exclusively for noise-related domain shift, adapting a model to a new speaker, a new recording setup, or a new language, rather than to a genuinely different kind of distortion. Whether these methods still work when the distortion itself changes, from background noise to reverberation, has not been studied before this research work.

Reverberation, the second major class of speech distortion studied in this project, arises from a fundamentally different physical process than additive noise [6, 7].

Denoising addresses an additive corruption,

$$y(t) = x(t) + n(t), \tag{1.1}$$

where $y(t)$ is the observed, distorted speech signal, $x(t)$ is the underlying clean speech, and $n(t)$ is the additive background noise. Dereverberation instead addresses a convolutive corruption,

$$y(t) = x(t) * h(t), \tag{1.2}$$

where $h(t)$ is a room impulse response (RIR) and $*$ denotes convolution. This distinction between an additive and a convolutive corruption has a direct consequence for adaptation methods, developed in full in Chapter 5: a method that works by estimating and subtracting an additive component, or by assuming speech and noise occupy separable regions of a signal, relies on a structure that a convolutive distortion does not have. In the additive case, the noise $n(t)$ is a separate signal simply summed on top of the clean speech: the clean signal's own structure is untouched, and in principle the noise could be identified and subtracted away as an independent component. In the convolutive case, there is no separate signal to subtract. The room impulse response $h(t)$ describes how the speech signal itself gets smeared across time, as the same sound arrives at a microphone repeatedly via direct path and reflections off walls, ceiling, and objects in the room, each delayed and attenuated differently. Reverberation is therefore a transformation applied to the speech signal, not an addition to it. This report returns to this distinction repeatedly, since it turns out to explain both why the TTA methods evaluated here struggle specifically with reverberation, and, as developed in Chapter 5, why the most effective way found to adapt a model to reverberation was to not adapt it directly to reverberation at all.

This project's first research question below is a direct consequence of this additive/convolutive distinction: since existing TTA methods were built and validated around the additive case, whether they transfer to the convolutive case is not something that could be assumed and had to be tested.

1.2 Research Questions

This research work investigates test-time adaptation for speech enhancement beyond the noise-related setting it has predominantly been studied in. Three questions structure the investigation:

1. Do existing TTA methods (MPol [8], LaDen [9], RemixIT [10]), built for noise-related domain shift, transfer directly to reverberation, and if not, why do they fail?
2. Does training a model jointly on both distortion types change how it responds to TTA, compared to a model trained on only one distortion type?
3. If a model has been adapted to one distortion type, does that adaptation transfer to a different, unadapted distortion type, even without any further adaptation on that second type?

The third question turned out to motivate the central finding of this research work, described in full in Chapter 4 and analyzed in Chapter 5. Stated plainly, ahead of the detailed results: **the most effective way found to adapt a model to reverberation was to not adapt it to reverberation directly at all**, but instead to adapt it on noise-related data and rely on the resulting improvement transferring across to reverberation on its own.

1.3 Contributions

This project makes the following contributions:

- Four CMGAN models are trained under controlled conditions that isolate the effect of task composition: one trained only on denoising, one trained only on dereverberation, and two trained jointly on both tasks using different underlying noise corpora (Chapter 3).
- Three existing TTA methods (MPol, LaDen, RemixIT) and one project-specific reverb-oriented variant (RIRmixIT) are evaluated across both distortion types, both in isolation and in combination, establishing that all denoising-oriented methods degrade performance when applied directly to reverberation (Section 4.5).
- A cross-task transfer effect is identified and confirmed across multiple TTA methods, multiple datasets, and both directions of transfer (denoising to reverberation and reverberation to denoising): adapting a model on one distortion type and evaluating it directly on the other, without further adaptation, outperforms adapting directly on the target distortion type (Section 4.9).
- A mechanistic explanation for this transfer effect is developed and supported with a weight-level analysis, showing that TTA concentrates its updates in the normalization statistics of the model's shared conformer blocks rather than in task-specific learned weights (Section 5.2).
- Five hypotheses (H1 through H5) are formulated, tested against the experimental results, and where possible reconciled with one another, providing a structured account of when and why TTA helps or hurts under each distortion type (Chapter 5).

1.4 Report Structure

The remainder of the report is organized as follows. Chapter 2 lays the technical groundwork: the CMGAN architecture, the idea of test-time adaptation and how it has previously been formulated, the specific TTA methods used here, and the metrics used to judge enhancement quality for both denoising and dereverberation. Chapter 3 then walks through the datasets, model variants, and adaptation configurations behind the experiments. The experimental results themselves, across every phase of the investigation, are presented in Chapter 4, and Chapter 5 works through what those results mean, developing the hypotheses that tie them together. The report closes with Chapter 6, which summarizes the findings, is honest about where the project falls short, and points to what should be done next.

2 Background

2.1 Types of Distortion

A distorted speech signal observed in a real acoustic environment can be modeled as

$$y(t) = x(t) * h(t) + n(t), \tag{2.1}$$

where $y(t)$ is the observed, distorted speech signal, $x(t)$ is the underlying clean speech, $n(t)$ is additive background noise, $h(t)$ is a convolutive filter representing the acoustic path from source to microphone, and $*$ denotes convolution. Different speech enhancement tasks isolate different terms of this equation. Denoising addresses the additive term alone, taking $h(t)$ to be an identity filter and treating $n(t)$ as the sole corruption to remove. Dereverberation instead treats $h(t)$ as a room impulse response (RIR), the convolutive signature of reflections off walls and objects in a room, and aims to invert or suppress its effect. A third task, bandwidth extension or super-resolution, treats $h(t)$ as a low-pass filter representing missing high-frequency content, and is not used in this project.

This project's four base models each correspond to a different combination of the additive and convolutive terms above: a model trained only on denoising data, a model trained only on dereverberation data, and two models trained jointly on both. As introduced in Section 1.1, the distinction between additive and convolutive distortion is not merely notational: it determines which adaptation strategies can work at all, since the assumptions built into existing test-time adaptation methods for speech enhancement are, in each case, assumptions that hold for the additive case and do not hold for the convolutive one.

2.2 CMGAN

All four base models in this project use the CMGAN architecture (Conformer-based Metric Generative Adversarial Network), originally proposed for denoising [1] and later extended to cover dereverberation and super-resolution [2]. CMGAN operates on the complex short-time Fourier transform (STFT) representation of the input signal rather than on the raw waveform or magnitude spectrogram alone. The STFT converts a waveform into a time-frequency representation, breaking the signal into short overlapping windows and reporting, for each window, how much energy is present at each frequency, along with the phase (timing offset) of that energy. Working with the complex STFT, magnitude and phase together, allows CMGAN to model phase information directly rather than reusing the noisy input's phase unmodified, which matters because phase errors are a common source of artifacts in enhanced speech.

The architecture consists of a shared encoder, built from an initial convolution block followed by a dilated DenseNet with dilation factors $\{1, 2, 4, 8\}$ (each successive layer skips over a wider spacing of input positions, which widens the range of context the network can see without adding parameters or depth), followed by four Two-Stage Conformer Blocks (TSCB). Each TSCB applies two sequential conformer blocks, one attending across the time axis and one across the frequency axis, combining the long-range modeling capacity of self-attention, a mechanism that lets the model directly relate distant positions in a sequence to one another, with the local feature extraction of convolution [11]. The encoder output feeds two parallel, structurally identical decoder branches: a mask decoder, which predicts a time-frequency mask multiplicatively applied to the input magnitude, and a complex decoder, which predicts a direct additive correction to the real and imaginary spectral components. This decoupled mask-and-complex-decoder design is intended to resolve a known limitation of purely mask-based approaches, where imperfect phase reconstruction limits achievable quality regardless of how accurate the magnitude mask is. Figure 2.1 below shows this full pipeline; the TSCB blocks are worth noting in particular, since they are the component referenced throughout Chapter 5's discussion of H1, where their normalization statistics are shown to be what test-time adaptation actually updates.

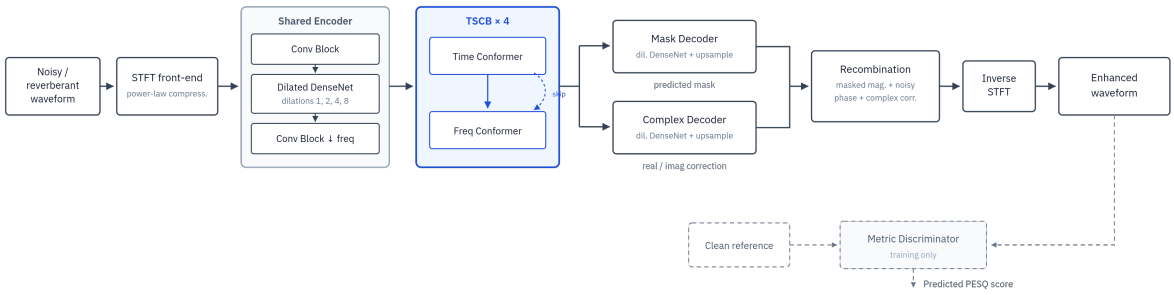


Figure 2.1: **CMGAN Architecture.** STFT front-end, dilated DenseNet encoder, four TSCB blocks, and the decoupled mask/complex decoder branches.

CMGAN is trained adversarially against a metric discriminator, following the MetricGAN framework [3, 4], in which the discriminator is trained to predict the PESQ score of a given output rather than to simply distinguish real from generated speech. This provides a differentiable training signal correlated with the non-differentiable evaluation metric ultimately used to judge output quality.

With the architecture that every model in this project is built on now established, the next question is what happens when this architecture meets conditions it was not trained on, and what can be done about it.

2.3 Domain Shift

Speech enhancement models are trained on a fixed, finite set of recording conditions. A deployed system, however, has to cope with whatever acoustic environment, speaker, and noise type it happens to encounter, and these rarely match the training data exactly [8]. This mismatch is called domain shift, and its effect on performance is not subtle: a model trained under one set of conditions measurably performs worse once those conditions change [8].

The difficulty is not that training sets are too small, but that the space of conditions a system might face after deployment cannot be enumerated in advance. No matter how large or diverse a training set is, it cannot include every room, every microphone, and every noise source a model will eventually run into. One could try to close this gap by collecting labeled data for each new condition as it arises, but that is rarely practical: recordings paired with a clean reference are expensive to obtain, and are usually unavailable at deployment time in the first place.

Two established paradigms respond to this problem, each with a different cost. Supervised finetuning solves it directly, but needs exactly the labeled target-domain data that is typically missing. Unsupervised domain adaptation (UDA) drops the label requirement, yet still assumes access to the original source training data during adaptation [12, 13], and that assumption carries its own cost: retaining or sharing source data raises privacy concerns and storage overhead, and is not always even possible once a model has shipped [8]. Test-time adaptation, the subject of the next section, sidesteps both problems: it needs neither labels nor the original training data, and it adapts online, during inference itself, rather than as a separate offline step with its own latency cost [8].

2.4 Test-Time Adaptation

Test-time adaptation is one way to close the gap just described: a way to update a trained model for a new acoustic condition without retraining it from scratch.

Test-time adaptation (TTA) is an online adaptation paradigm in which a trained model updates a subset of its parameters using only the unlabeled data it encounters at deployment time, with adaptation and inference occurring in the same pass and with no access to the model's original training data [5, 14]. This distinguishes TTA from two related but less constrained paradigms. Supervised finetuning assumes access to labeled data from the target domain. Unsupervised domain adaptation (UDA) assumes access to the source training data, even if the target data remains unlabeled, and is typically performed offline before deployment [12, 13]. TTA assumes neither.

TTA was first developed for image classification, where a natural adaptation signal is available: a well-calibrated classifier should produce confident (low-entropy) predictions on in-distribution data. Entropy here refers to how spread out or uncertain a model's predicted probabilities are across the possible classes; a confident prediction concentrates most of its probability on one class (low entropy), while an uncertain prediction spreads probability more evenly across several classes (high entropy). Since entropy tends to rise under distribution shift, minimizing prediction entropy at test time provides a self-supervised adaptation objective [5]. This entropy-based signal does not transfer to speech enhancement, since SE models output a continuous signal estimate rather than a categorical prediction, and there is no analogous entropy to minimize. Each TTA method used in this project therefore substitutes a different, domain-specific self-supervised signal in place of entropy minimization.

A distinction that recurs throughout the rest of this report is between adapting a model directly on the domain it is then evaluated on, and adapting it on one domain while evaluating on another, with or without a further adaptation step on that second domain. This distinction underlies the central results in Sections 4.9 and 4.8, and the H3/H4 discussion in Section 5.4.

The description above establishes what TTA is and why it differs from other adaptation paradigms. What it has not yet established is the actual mechanics of a single adaptation step: which parameters change, when the loss is computed, and how the gradient reaches those parameters. This mechanism is shared by all four methods evaluated in this project, even though, as detailed method by method in Section 2.5, the four differ substantially in what additional machinery they require to produce their loss signal.

Let θ denote the full set of parameters of a trained CMGAN model, and let $\theta_{\text{adapt}} \subset \theta$ denote the small subset of those parameters, restricted to normalization and output layers, that any given TTA method is permitted to update, following the restriction established in the MPol and LaDen papers and applied identically to all four methods in this project for comparability (Chapter 3 details the exact parameter count). At each inference step t, the model receives a new, unlabeled input y_t from the target domain and produces an output $\hat{x}_t = f_\theta(y_t)$ in the same forward pass used for ordinary inference. A method-specific self-supervised loss $\mathcal{L}_t$ is then computed from this same forward pass, using no labeled target-domain data, only quantities the method can derive from y_t, $\hat{x}_t$, and, depending on the method, any additional signal derived from these two (Section 2.5 specifies each method's $\mathcal{L}_t$ individually). The adaptable parameters are then updated by a single gradient step,

$$\theta_{\text{adapt}} \leftarrow \theta_{\text{adapt}} - \eta \nabla_{\theta_{\text{adapt}}} \mathcal{L}_t, \tag{2.2}$$

where η is the adaptation learning rate, while every parameter outside θ_{adapt} remains exactly as it was after the model's original training and receives no gradient at all. This loop, computing $\hat{x}_t$, computing $\mathcal{L}_t$ from that same forward pass, then updating only θ_{adapt}, repeats independently at every subsequent inference step, with no separate offline phase and no access to any input seen before step t beyond whatever has already been folded into θ_{adapt} by earlier updates.

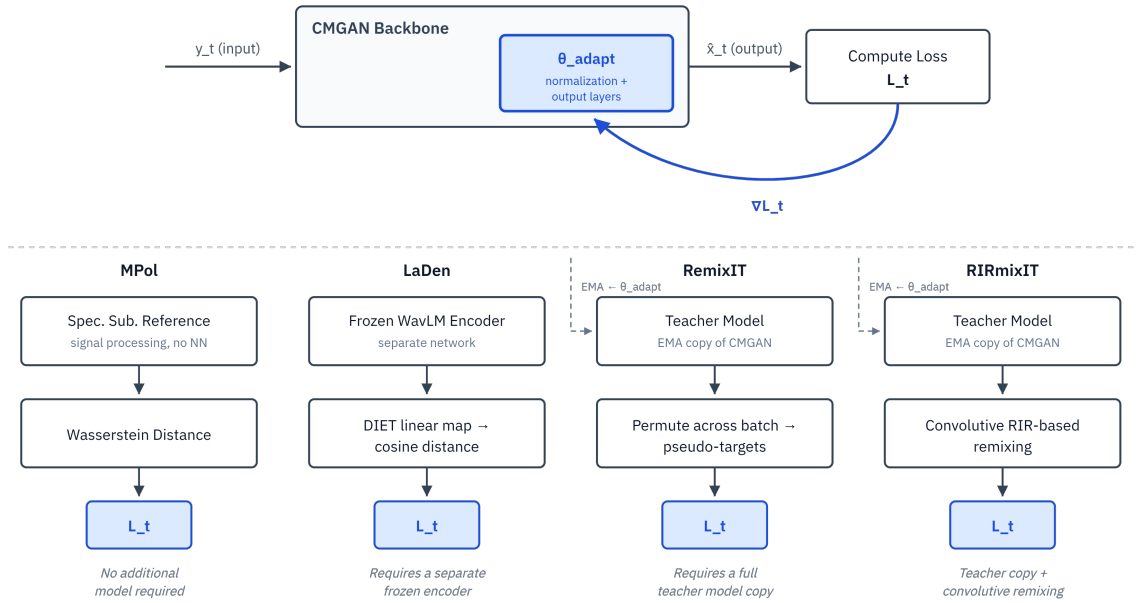


Figure 2.2: **Shared Adaptation Mechanism.** The update loop and each method's extra machinery, from none (MPol) to a full teacher copy (RemixIT/RIRmixIT).

2.5 TTA Methods

Four TTA methods are evaluated in this project.

MPol (Mask Polarization) [8] is based on the observation that a well-calibrated mask-based SE model should produce a bimodal mask, with most time-frequency bins confidently assigned to either speech or noise. Under domain shift, this bimodality tends to collapse. MPol restores it by matching the distribution of the model's predicted mask against a reference mask constructed via spectral subtraction, using the Wasserstein distance, a measure of how much probability mass would need to be moved, and how far, to reshape one distribution into another, between the two distributions as the adaptation loss. Architecturally, MPol requires no additional model beyond CMGAN itself: both the predicted mask and the spectral-subtraction reference mask are computed directly from CMGAN's own forward pass and simple signal processing, with no separate learned component involved in producing the loss.

LaDen (Latent Denoising) [9] is based on the hypothesis that the relationship between clean and noisy speech, complex at the raw signal level, becomes approximately linear in the embedding space of a large pretrained speech encoder. This claim is easiest to understand by analogy. At the raw audio level, removing noise from speech is a complicated, highly nonlinear problem: there is no simple arithmetic operation that reliably turns a noisy waveform back into a clean one. A pretrained speech encoder, however, is trained to compress raw audio into a compact numerical representation, or embedding, that discards much of the signal's surface-level complexity while preserving the information relevant to the encoder's training objective. LaDen's finding is that once both the clean and the noisy version of an utterance are projected into this embedding space, the relationship between the two projected points

is well approximated by a single linear transformation, effectively a matrix multiplication, even though no such simple relationship holds between the original waveforms. This is comparable to a problem that looks intractable in one coordinate system becoming simple once viewed from a better-chosen perspective. A linear transformation matrix implementing this mapping is estimated offline on labeled source data using a frozen WavLM encoder [15], and used at test time to produce a pseudo-target for the model's output in embedding space, without requiring any labels at adaptation time. Architecturally, this makes LaDen heavier than MPol: at every adaptation step, both the noisy input and CMGAN's own output must additionally be passed through this separate, frozen WavLM CNN encoder [15], a distinct 4.2-million-parameter network run alongside CMGAN purely to produce the loss signal, adding meaningful compute overhead relative to MPol despite updating the same narrow set of CMGAN parameters. This linearity claim, and specifically whether it continues to hold when the distortion is convolutive reverberation rather than additive noise, is central to the H5 hypothesis discussed in Section 5.5.

RemixIT [10] assumes the target domain's distortion is additive and separable. A teacher model produces estimated speech and noise components for a batch of target-domain data; these components are permuted across the batch and remixed into new bootstrapped mixtures with corresponding pseudo-targets, which a student model is then trained to reconstruct. Architecturally, this requires maintaining a full second copy of the CMGAN model, the teacher, alongside the student model actually being adapted; the teacher's weights are periodically updated from the student via an exponential moving average, rather than being trained directly, but the memory and compute cost of running two full CMGAN forward passes per adaptation step is substantially heavier than either MPol's or LaDen's machinery.

RIRmixIT is a project-internal variant of RemixIT, developed for this project specifically to test whether a method built around a convolutive rather than additive distortion model would fare better under dereverberation domain shift. It has no independent published citation and should be understood as an internal adaptation of RemixIT's bootstrapped-remixing framework, not as an established method from the literature. Architecturally, RIRmixIT carries the same teacher-copy overhead as RemixIT; the only difference is that the remixing step inside the teacher branch constructs its bootstrapped mixtures via convolutive, RIR-based combination rather than additive permutation. This is a direct application of the additive/convolutive distinction discussed in Section 1.1: RemixIT's additive permutation assumes speech and noise can be separated and recombined as independent components, an assumption that does not hold for reverberation, so RIRmixIT replaces that step with a convolutive construction instead, while retaining RemixIT's full computational footprint.

All four methods restrict their parameter updates to normalization and output layers of the underlying model, a design choice explicit in the MPol and LaDen papers and applied consistently to RemixIT and RIRmixIT in this project's implementation for comparability. As detailed above and shown in Figure 2.2, the four methods are not simply four different loss functions bolted onto identical machinery; they differ substantially in the additional components required to produce that loss, ranging from none (MPol) to a separate frozen encoder (LaDen) to a full second model copy (RemixIT, RIRmixIT), while all four ultimately converge on updating the same narrow subset of CMGAN's parameters. The consequences of this shared restriction, and its relationship to this project's findings on adaptation ordering, are discussed in Chapter 5. Each method's core assumption about the structure of the distortion it corrects, summarized for reference in Table 5.2, becomes the central explanatory

device for why direct reverberation TTA fails in Chapter 5's H2 discussion.

2.6 Evaluation Metrics

Speech enhancement quality is assessed with objective metrics that correlate with human perceptual judgment, since no single metric captures every aspect of quality. This project follows a task-specific metric split, using different metric sets for denoising and dereverberation evaluation, established in the predecessor project this project builds on [16].

For denoising, the primary metrics are PESQ [17], the ITU-T standard perceptual quality measure and the single most widely used SE metric; STOI [18, 19], which measures intelligibility rather than general quality; the composite measures CSIG, CBAK, and COVL [20], derived from regression against subjective ratings and targeting signal distortion, background intrusiveness, and overall quality respectively; and segmental and frequency-weighted segmental SNR (SSNR, FWSEG) [21].

Segmental SNR is computed by averaging the signal-to-noise ratio over short frames rather than over the whole signal at once, which makes it more sensitive to localized noise bursts than a single global SNR value:

$$\text{SSNR} = \frac{1}{M}\sum_{m=1}^{M} 10\log_{10}\frac{\sum_n x^2(n,m)}{\sum_n \left(x(n,m)-\hat{x}(n,m)\right)^2}, \tag{2.3}$$

where M is the number of frames, $x(n,m)$ is the clean reference signal at sample n within frame m, and $\hat{x}(n,m)$ is the enhanced signal at the same position. FWSEG extends this by additionally weighting each frequency band according to its perceptual importance to intelligibility before averaging.

SI-SDR measures signal distortion in a way that is invariant to a global scaling mismatch between the estimate and the reference:

$$\text{SI-SDR} = 10\log_{10}\frac{\|\alpha x\|^2}{\|\alpha x-\hat{x}\|^2}, \qquad \alpha = \frac{\hat{x}^\top x}{\|x\|^2}, \tag{2.4}$$

where x is the clean reference signal, $\hat{x}$ is the enhanced signal, and α is the scalar that projects the enhanced signal onto the reference, removing any overall amplitude difference before the distortion is measured.

For dereverberation, the primary metrics are PESQ and SI-SDR, shared with the denoising set; FWSEG; the log-likelihood ratio (LLR) [22], a measure of spectral envelope similarity where lower values indicate better dereverberation; cepstral distance (CD) [23], a related measure in the cepstral domain; and the speech-to-reverberation modulation energy ratio (SRMR) [24].

LLR compares the linear predictive coding (LPC) spectral envelopes of the clean and enhanced signals:

$$\text{LLR} = \log\frac{a_{\hat{x}}^\top R_x\, a_{\hat{x}}}{a_x^\top R_x\, a_x}, \tag{2.5}$$

where a_x and $a_{\hat{x}}$ are the LPC coefficient vectors of the clean and enhanced signals respectively, and R_x is the autocorrelation matrix of the clean signal; a smaller LLR indicates the

enhanced signal's spectral envelope more closely matches the clean signal's. CD is computed analogously, but in the cepstral domain rather than directly on LPC coefficients, comparing the cepstral coefficients derived from each signal's spectral envelope.

SRMR does not compare against a clean reference at all. It instead computes the ratio of modulation energy at low modulation frequencies, associated with speech, to modulation energy at high modulation frequencies, associated with reverberation, within an auditory-inspired filterbank decomposition of the signal:

$$\text{SRMR} = \frac{\sum_{k=1}^{4} E_k}{\sum_{k=5}^{8} E_k}, \tag{2.6}$$

where E_k is the modulation energy in the k-th modulation frequency band of an 8-band decomposition, with lower-indexed bands corresponding to slower modulation frequencies typical of speech and higher-indexed bands corresponding to faster modulation typical of reverberant reflections. A higher SRMR indicates less reverberation relative to speech energy.

SRMR is the only non-intrusive metric used in this project, meaning it requires no clean reference signal to compute, visible directly in its formula above by the absence of any reference-signal term. This distinction matters specifically for TTA: at genuine deployment time, no clean reference is available, so SRMR is the only metric in this project's evaluation suite that reflects what could, in principle, actually be measured and used as an adaptation signal under real deployment conditions. Every other metric is intrusive and available only in the controlled research setting used throughout this project's experiments.

3 Experimental Setup

3.1 Motivation

The choices behind this setup, which models were trained, on which data, and under which adaptation conditions, were made to keep the comparisons in the following chapters as fair as possible. Each design decision described below follows from that goal.

3.2 Datasets

3.2.1 Clean Speech

EARS provides 100 hours of clean, anechoic speech from 107 speakers, and contains no noise or reverberation of its own [25]. It is the clean speech source for every training set used in this research work.

The clean speech behind the VoiceBank evaluation set comes from the Voice Bank corpus [26], a collection of read English sentences recorded across a range of regional accents. The specific evaluation set used here is the standard 30-speaker Voice Bank benchmark widely used in speech enhancement research, split into 28 training speakers and 2 held-out test speakers, recorded at 48kHz and later downsampled for this project's pipeline.

The clean speech behind the DNS English evaluation set is sourced from LibriVox, a repository of public-domain audiobook narrations, and spans several thousand speakers reading in English [27]. Unlike EARS or Voice Bank, which were recorded under controlled studio conditions specifically for speech research, LibriVox recordings vary considerably in recording quality, since they were produced by volunteer narrators using their own equipment; the DNS Challenge organizers filter this pool for acceptable quality before release.

3.2.2 Denoising Datasets

Four datasets are used for denoising: EARSWHAM, EARS_DEMAND, VoiceBank, and DNS English. EARSWHAM mixes EARS speech with real noise recordings from the WHAM! dataset [28], following the construction defined by the EARS dataset's own authors. EARS_DEMAND mixes EARS speech with noise recordings from the DEMAND dataset [29] instead, using the same construction with a different noise source. VoiceBank, whose clean speech is described above, is paired with DEMAND noise following the standard construction for this benchmark. DNS English, whose clean speech is also described above, is paired with the noise conditions defined by the DNS Challenge itself.

3.2.3 Reverberation Datasets

Three datasets are used for reverberation: EARSRev, REVERB Challenge [7], and EARS+ACE. EARSRev [25] convolves EARS speech with a collection of real recorded room impulse responses, following the construction defined by the EARS dataset's own authors, producing reverberant speech with no additive noise involved. REVERB Challenge is a separate reverberant speech corpus, combining real and simulated recordings under different room and speaker conditions. EARS+ACE is synthetic reverberant speech built by convolving EARS speech with room impulse responses from the ACE challenge [30], keeping the same speaker population as EARS but with a different set of room impulse responses than EARSRev.

3.2.4 Dataset Usage

EARSWHAM and EARS_DEMAND are each used to train a model on denoising alone. EARSRev is used to train a model on reverberation alone. EARSCombined v1 interleaves EARSRev with EARS_DEMAND, and EARSCombined v2 interleaves EARSRev with EARSWHAM, each at a fifty-fifty ratio between the two distortion types; these are used to train the two jointly-trained model variants described in the next section.

VoiceBank, DNS English, REVERB Challenge, and EARS+ACE are not used for training. They are held out for evaluation, to test how a model trained on the datasets above performs under conditions it has not seen. VoiceBank introduces a speaker and noise shift relative to the training data. DNS English introduces a language and noise shift. REVERB Challenge introduces a room and speaker shift under reverberant conditions. EARS+ACE keeps the same speaker population as the training data and introduces only a shift in the reverberation condition itself.

3.3 Infrastructure

All training and adaptation experiments were run on a SLURM-managed compute cluster (SLURM being a common job scheduler used to allocate compute resources on shared high-performance computing, or HPC, clusters) using NVIDIA RTX A5000 GPUs (24GB), with a maximum job wall-clock limit of seven days per job. The codebase, evaluation pipeline, and all TTA method implementations are maintained in a version-controlled repository (`https://github.com/rustombhesania/SpeechEnhancement`), with checkpoints, adaptation matrices, and experiment configurations stored on cluster-attached network storage. All software dependencies were managed through a dedicated Python environment specific to this project.

3.4 Model Variants

Four CMGAN models are trained. Two are trained on a single distortion type in isolation. Two are trained on a mix of EARSRev with EARSWHAM or EARS_DEMAND, interleaved

at a fifty-fifty ratio between the two distortion types. All four are trained for 25 epochs. Isolated Denoising and Mixed v2 are trained a second time, to 50 epochs, to compare the effect of training duration. Table 3.1 lists all four models.

Table 3.1: Model variants.

Model	Training Data	Distortion Type	Epochs
Isolated Denoising	EARSWHAM	Denoising only	25, 50
Isolated Reverb	EARSRev	Reverberation only	25
Mixed v1	EARSRev + EARS_DEMAND	Both, jointly trained	25
Mixed v2	EARSRev + EARSWHAM	Both, jointly trained	25, 50

4 Results

4.1 TTA Configuration

All four TTA methods described in Section 2.5 are applied under a shared adaptation budget: only normalization and output layers of the generator are updated, corresponding to approximately 6,020 of the model's 2,016,483 total parameters, roughly 0.3% of the model. This restriction follows the design of the MPol and LaDen methods as published, and is applied identically to RemixIT and RIRmixIT in this project's implementation to keep all four methods directly comparable.

Each experiment adapts a given model on a target dataset for a single online epoch. In this context, an epoch refers to one full pass through the target adaptation dataset during which the model is simultaneously being updated and evaluated, distinct from a standard training epoch, where the model is only being updated and evaluation happens separately afterward. Using a single online epoch is consistent with the TTA paradigm's requirement that adaptation and inference occur together rather than as a separate offline phase. Where an experiment involves adapting on two datasets in sequence (Section 4.8), the model checkpoint produced by the first adaptation step is carried forward as the starting point for the second step, with no reset to the original source weights in between. This absence of a reset is a deliberate methodological choice rather than a default: it is precisely what makes the second adaptation step's effect measurable as a modification of the first step's result, and it is what the H3 finding in Chapter 5 (that this second step consistently hurts) is a direct statement about.

Throughout this chapter, bold values indicate the best score per metric within each table, restricted to tables comparing multiple methods against a common source baseline on the same evaluation condition. Tables comparing different evaluation conditions, datasets, or configurations against one another, rather than comparing methods head to head, are not bolded, since no single value in those tables represents a "best" result in a comparable sense.

4.2 In-Distribution Baselines

Before evaluating any domain shift or adaptation, each of the four CMGAN models was evaluated on its own training distribution to establish an in-distribution baseline. Table 4.1 shows the isolated denoising model, evaluated on EARSWHAM at both 25 and 50 epochs.

Table 4.1: Isolated Denoising model (trained on EARSWHAM), in-distribution.

Epochs	PESQ↑	STOI↑	CSIG↑	CBAK↑	COVL↑	SSNR↑	FWSEG↑	SI-SDR↑
25	2.337	0.822	3.590	2.950	2.946	6.490	10.159	9.613
50	2.388	0.826	3.637	2.980	2.998	6.428	10.029	9.043

The isolated reverb model, evaluated the same way on EARSRev, is shown in Table 4.2.

Table 4.2: Isolated Reverb model (trained on EARSRev), in-distribution.

Epochs	PESQ↑	SI-SDR↑	FWSEG↑	LLR↓	CD↓	SRMR↑
25	1.982	-8.555	6.409	0.437	3.960	6.995
50	2.070	-9.850	6.372	0.437	4.018	6.836

Finally, Table 4.3 shows both mixed models, with routing active so that reverb metrics are averaged only over reverb segments and denoising metrics only over denoising segments.

Table 4.3: Mixed models, in-distribution, routing active. Reverb metrics averaged over reverb segments only, denoising metrics over denoising segments only.

Model	Epochs	PESQ↑	STOI↑	CSIG↑	CBAK↑	COVL↑	SSNR↑	FWSEG↑	SI-SDR↑	LLR↓	CD↓	SRMR↑
Mixed v1	25	2.331	0.868	3.774	2.482	3.043	-1.250	8.063	-1.040	0.518	3.823	7.234
Mixed v2	25	2.140	0.850	3.557	2.497	2.831	0.735	8.158	-0.697	0.608	4.031	7.753
Mixed v2	50	2.228	0.855	3.625	2.536	2.912	0.582	7.996	-1.071	0.613	3.953	7.872

4.3 Domain Shift Without TTA

To establish how much performance degrades under distribution shift before any adaptation, the isolated models were evaluated on out-of-distribution datasets with no TTA applied.

Notably, PESQ *improves* under both shifts relative to the in-distribution EARSWHAM baseline, suggesting VoiceBank+DEMAND and DNS present an easier denoising task for this model than its own training distribution.

Unlike the denoising model, the reverb model's PESQ drops under domain shift, consistent with reverb being a harder generalization problem across room conditions.

Table 4.4: Isolated Denoising model (25 epochs, trained on EARSWHAM), domain shift without TTA.

Dataset	Shift Type	PESQ↑	STOI↑	CSIG↑	CBAK↑	COVL↑	SSNR↑	FWSEG↑
EARSWHAM	In-distribution	2.337	0.822	3.590	2.950	2.946	6.490	10.159
VoiceBank+DEMAND	Speaker + noise shift	2.624	0.941	3.846	2.744	3.189	2.358	11.590
DNS (English)	Language + noise shift	2.369	0.923	3.622	2.982	2.986	6.781	10.964

Table 4.5: Isolated Reverb model (25 epochs, trained on EARSRev), domain shift without TTA.

Dataset	Shift Type	PESQ↑	SI-SDR↑	FWSEG↑	LLR↓	CD↓	SRMR↑
EARSRev	In-distribution	1.982	-8.555	6.409	0.437	3.960	6.995
REVERB Challenge	Room + speaker shift	1.892	-6.867	5.796	0.490	4.314	5.824

4.4 Single-Task Denoising TTA

MPol, LaDen, and RemixIT were each applied to the isolated denoising model on VoiceBank+DEMAND and DNS. Two distinct kinds of metric appear throughout this chapter and are worth distinguishing explicitly before the results tables begin. Each TTA method optimizes its own internal adaptation loss during the adaptation process itself (for example, MPol's Wasserstein distance between predicted and reference masks, described in Chapter 2); this internal loss is never reported directly in this chapter's tables. Separately, every result reported below is an evaluation metric (PESQ, SI-SDR, and so on), computed against a clean reference after adaptation is complete, independent of whatever internal loss the method was actually optimizing. A method can have a well-behaved internal loss and still perform poorly on evaluation metrics, or vice versa; Chapter 5's H2 discussion relies on exactly this distinction, using each method's internal adaptation loss as diagnostic evidence for why its evaluation-metric performance degrades under reverberation.

Table 4.6: Single-task denoising TTA, isolated denoising model, VoiceBank+DEMAND.

Method	PESQ↑	CSIG↑	CBAK↑	COVL↑	SSNR↑	SI-SDR↑
Source	2.624	3.846	2.744	3.189	2.358	8.444
MPol	2.612	3.831	2.733	3.175	2.292	8.471
LaDen	2.477	3.795	2.743	3.096	**3.110**	**9.750**
RemixIT	**2.625**	**3.850**	**2.759**	**3.191**	2.601	8.420

Of the three, RemixIT comes out as the strongest perceptual method, matching or beating source on PESQ, CSIG, CBAK, and COVL across both datasets. LaDen takes a different path, trading some PESQ for gains in SI-SDR and SSNR, while MPol stays closest to source throughout. None of the gains are large, however, and the reason is not hard to find: the isolated denoising model was already well-adapted to both shift datasets (Section 4.3), so there was not much headroom left for TTA to exploit in the first place. Section 4.13 revisits this and confirms it is not specific to VoiceBank.

Table 4.7: Single-task denoising TTA, isolated denoising model, DNS (English).

Method	PESQ↑	CSIG↑	CBAK↑	COVL↑	SSNR↑	SI-SDR↑
Source	**2.369**	**3.622**	**2.982**	**2.986**	**6.781**	10.510
MPol	2.350	3.583	2.967	2.955	6.756	10.540
LaDen	2.309	3.542	2.921	2.910	6.377	10.500
RemixIT	2.367	3.603	2.973	2.973	6.736	**10.689**

4.5 Single-Task Reverb TTA

The same three methods, plus RIRmixIT, were applied to the isolated reverb model on REVERB Challenge.

Table 4.8: Single-task reverb TTA, isolated reverb model, REVERB Challenge.

Method	PESQ↑	SI-SDR↑	FWSEG↑	LLR↓	CD↓	SRMR↑
Source	**1.892**	-6.867	**5.796**	**0.490**	**4.314**	**5.824**
MPol	1.665	-6.617	5.014	0.544	4.809	5.436
LaDen	1.609	-6.842	5.085	0.566	4.806	5.082
RemixIT	1.563	-9.512	4.752	0.536	4.792	5.103
RIRmixIT	1.708	**-6.466**	4.951	0.553	4.750	4.638

Core negative finding of this chapter: every TTA method hurts reverb performance, and on every metric, relative to source. RemixIT fares worst, which fits with its additive-distortion assumption being the least appropriate of the four for convolutive reverb. RIRmixIT, built specifically for reverb, does better than the other three but still cannot close the gap to the untouched source model. This result is the starting point for the assumption-mismatch hypothesis developed in Chapter 5.

4.6 Multi-Task Simultaneous TTA

Both mixed models were adapted on EARSCombined, the same reverb+denoising interleaved distribution used in training, to test whether simultaneous multi-task TTA is stable.

Simultaneous adaptation turns out to be stable for both mixed models: unlike the isolated-reverb case in Section 4.5, no method causes a catastrophic collapse. RemixIT gives the most consistent improvement, particularly on CBAK and SSNR, while LaDen improves the reverb-specific metrics LLR and CD on both models. The gains are modest overall, which is not surprising given how little room this in-distribution setting leaves for improvement, in contrast to the out-of-distribution results that follow in Section 4.7.

Table 4.9: Multi-task simultaneous TTA, Mixed v1 (EARSRev + EARS_DEMAND), in-distribution.

Method	PESQ↑	STOI↑	CSIG↑	CBAK↑	COVL↑	SSNR↑	FWSEG↑	SI-SDR↑	LLR↓	CD↓	SRMR↑
Source	**2.331**	**0.868**	3.774	2.482	**3.043**	-1.250	8.063	**-1.040**	0.518	3.823	**7.234**
MPol	2.312	0.867	3.739	2.471	3.014	-1.239	8.124	-1.532	0.550	3.837	7.198
LaDen	2.306	0.867	**3.777**	2.523	3.032	-0.418	7.947	-1.741	**0.484**	3.771	6.957
RemixIT	2.308	0.865	3.767	**2.591**	3.026	**0.710**	**8.130**	-1.106	0.491	**3.764**	7.045

Table 4.10: Multi-task simultaneous TTA, Mixed v2 (EARSRev + EARSWHAM), in-distribution.

Method	PESQ↑	STOI↑	CSIG↑	CBAK↑	COVL↑	SSNR↑	FWSEG↑	SI-SDR↑	LLR↓	CD↓	SRMR↑
Source	2.140	**0.850**	3.557	2.497	2.831	0.735	8.158	-0.697	0.608	4.031	**7.753**
MPol	2.131	0.850	3.541	2.478	2.817	0.575	8.182	-0.895	0.612	4.023	7.605
LaDen	2.114	0.850	**3.562**	2.494	2.820	0.928	7.894	-1.004	**0.547**	4.010	7.345
RemixIT	**2.144**	0.850	3.558	**2.552**	**2.832**	**1.652**	**8.199**	**-0.742**	0.607	**3.945**	7.407

4.7 Out-of-Distribution Denoising TTA on Mixed Models

The mixed models were then adapted on datasets outside their training distribution, VoiceBank+DEMAND (denoising shift) and REVERB Challenge (reverb shift), to test true domain-shift TTA rather than the in-distribution setting of Section 4.6.

Table 4.11: OOD denoising TTA on mixed models, VoiceBank+DEMAND (denoising shift).

Model	Method	PESQ↑	CSIG↑	CBAK↑	COVL↑	SSNR↑	SI-SDR↑
Mixed v1	Source	2.331	3.774	2.482	3.043	-1.250	-1.040
Mixed v1	MPol	2.490	3.841	2.917	3.157	4.461	8.661
Mixed v1	LaDen	2.373	3.770	2.866	3.062	4.511	8.211
Mixed v1	RemixIT	**2.497**	**3.870**	**2.955**	**3.176**	**4.976**	**8.725**
Mixed v2	Source	2.140	3.557	2.497	2.831	0.735	-0.697
Mixed v2	MPol	2.624	**3.894**	2.755	**3.222**	2.119	6.663
Mixed v2	LaDen	2.477	3.855	**2.773**	3.136	**3.169**	**8.104**
Mixed v2	RemixIT	**2.626**	3.886	2.768	3.218	2.369	6.394

This is the strongest single-task TTA result anywhere in this project: all three methods produce large gains on both mixed models, with Mixed v2 under RemixIT rising from PESQ 2.14 to 2.63. RemixIT comes out best on Mixed v1, while MPol and RemixIT are roughly comparable on Mixed v2. The mixed models clearly have substantial headroom for adaptation under denoising shift, in sharp contrast to the isolated denoising model in Section 4.4, which had almost none. The same models, adapted instead on REVERB Challenge to test reverb-domain shift, are shown in Table 4.12.

The picture on reverb is entirely different from VoiceBank+DEMAND: there is no clear improvement anywhere, and RemixIT's SI-SDR collapses badly, from -10.9 to -14.4. This is the same pattern already seen in Section 4.5, now confirmed to extend to mixed models

Table 4.12: OOD denoising TTA on mixed models, REVERB Challenge (reverb shift).

Model	Method	PESQ↑	SI-SDR↑	FWSEG↑	LLR↓	CD↓	SRMR↑
Mixed v1	MPol	**1.842**	-6.878	6.027	**0.507**	4.363	**5.438**
Mixed v1	LaDen	1.835	**-6.571**	**6.286**	0.532	**4.261**	5.057
Mixed v1	RemixIT	1.797	-10.920	5.732	0.508	4.347	5.316
Mixed v2	MPol	1.799	-12.214	5.706	0.574	4.235	**6.827**
Mixed v2	LaDen	**1.800**	**-12.049**	**5.979**	**0.564**	4.128	6.128
Mixed v2	RemixIT	1.794	-14.444	5.705	0.570	**4.106**	6.599

as well: TTA struggles specifically with the reverb distortion type, regardless of whether the underlying model was trained in isolation or jointly.

4.8 Sequential TTA

MPol was applied sequentially, adapting on one task, then continuing to adapt the same (already-adapted) model on the other task, in both possible orderings.

Table 4.13: Sequential TTA (MPol), Mixed v1, both orderings.

Strategy	VoiceBank PESQ↑ (step)	REVERB PESQ↑ (step)
Denoise-first → Reverb	2.490 (step 1)	1.833 (step 2)
Reverb-first → Denoise	1.842 (step 1)	2.485 (step 2)

Table 4.14: Sequential TTA (MPol), Mixed v2, both orderings.

Strategy	VoiceBank PESQ↑ (step)	REVERB PESQ↑ (step)
Denoise-first → Reverb	2.625 (step 1)	1.803 (step 2)
Reverb-first → Denoise	1.799 (step 1)	2.602 (step 2)

Task ordering does not meaningfully affect final performance for either model: VoiceBank PESQ stays within a tight 2.48–2.63 range and REVERB PESQ within 1.80–1.84, regardless of which task is adapted first. The second adaptation step neither meaningfully helps nor hurts the gains achieved by the first step alone. This order-independence, together with the stronger finding in Section 4.9 that skipping the second step entirely is best, is discussed jointly in Chapter 5.

4.9 Cross-Task Transfer: Core Finding

This section presents the central experimental finding of this project: adapting on one distortion type and evaluating directly on the other, **without any adaptation step on the evaluation domain**, produces substantially larger gains than adapting directly on the target domain. The evidence for this finding is built up in five parts. Section 4.9's first two subsections establish the effect in both directions, adapting on denoising and evaluating on reverberation, then the reverse. The third subsection replicates the denoising-to-reverberation direction on a second denoising dataset, to rule out the effect being specific to one dataset pairing. The fourth subsection replicates it again using each model's own training-distribution denoising data rather than an out-of-distribution one, and additionally tests what happens when a second adaptation step is added back in. The fifth subsection tests one further variant of adding a second step, chaining two different TTA methods together rather than reusing the same method twice. Figure 4.1 below summarizes the core comparison, adapting on VoiceBank+DEMAND and evaluating on REVERB Challenge with no second step, for each of the three TTA methods, using data from Tables 4.8 and 4.15.

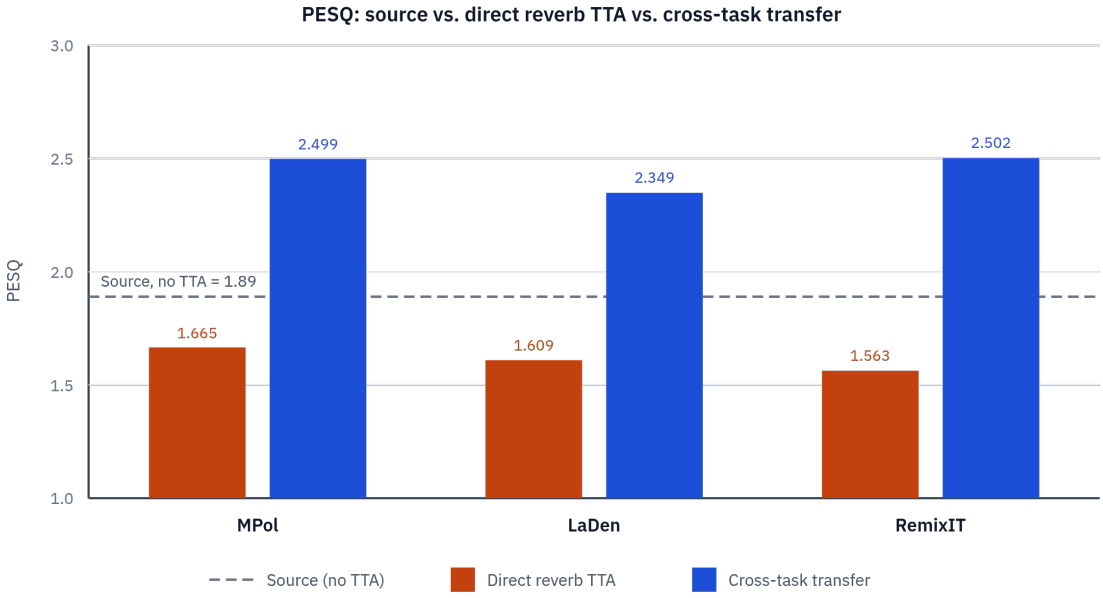


Figure 4.1: **Cross-Task Transfer Result.** PESQ under source, direct reverb TTA, and cross-task transfer, for each method.

4.9.1 Adapt Denoising (VoiceBank+DEMAND), Evaluate Reverb (REVERB Challenge)

PESQ rises from a source of 1.89 to 2.35–2.62 across all three methods and both models, well above even the best direct reverb TTA result from Section 4.5 (RIRmixIT, 1.71). SRMR, the only non-intrusive reverb metric and thus the metric most representative of true test-time conditions, rises from 5.82 to 8.18–9.34.

Table 4.15: Cross-task transfer: adapt on VoiceBank+DEMAND, evaluate on REVERB Challenge, no second adaptation step.

Model	Method	PESQ↑	SISDR↑	FWSEG↑	LLR↓	CD↓	SRMR↑
Source	(source)	1.892	-6.867	5.796	0.490	4.314	5.824
Mixed v1	MPol	2.499	8.640	**10.519**	0.553	3.817	8.345
Mixed v1	LaDen	2.349	7.709	10.503	**0.464**	**3.776**	8.179
Mixed v1	RemixIT	**2.502**	**8.830**	10.103	0.527	3.803	**8.466**
Mixed v2	MPol	**2.619**	6.615	10.345	0.556	3.516	**9.344**
Mixed v2	LaDen	2.421	**8.495**	**10.898**	**0.412**	3.668	8.664
Mixed v2	RemixIT	2.611	6.617	10.491	0.587	**3.506**	9.272

4.9.2 Adapt Reverb (REVERB Challenge), Evaluate Denoising (VoiceBank+DEMAND)

Table 4.16: Cross-task transfer: adapt on REVERB Challenge, evaluate on VoiceBank+DEMAND, no second adaptation step.

Model	Method	PESQ↑	STOI↑	CSIG↑	CBAK↑	COVL↑	SSNR↑	SISDR↑
Source Mixed v1	(source)	2.331	0.868	3.774	2.482	3.043	-1.250	-1.040
Source Mixed v2	(source)	2.140	0.850	3.557	2.497	2.831	0.735	-0.697
Mixed v1	MPol	**2.499**	**0.930**	3.810	**2.931**	**3.147**	**4.578**	**8.704**
Mixed v1	LaDen	2.482	0.927	**3.817**	2.899	3.136	4.425	7.927
Mixed v1	RemixIT	2.364	0.918	3.738	2.632	3.041	0.892	7.801
Mixed v2	MPol	**2.661**	**0.936**	**3.876**	**2.770**	**3.231**	**2.109**	**6.694**
Mixed v2	LaDen	2.603	0.929	3.838	2.634	3.179	0.551	3.096
Mixed v2	RemixIT	2.428	0.916	3.658	2.514	3.001	-0.046	3.990

The transfer holds in the reverse direction as well: adapting on REVERB Challenge and evaluating on VoiceBank+DEMAND (no denoising adaptation) yields PESQ 2.36–2.66, a large improvement over source (2.14–2.33). This reverse-direction result is not a minor confirmatory afterthought: together with the forward direction in Section 4.9's first subsection, it establishes that the transfer effect is genuinely bidirectional rather than an artifact of one particular distortion type being easier to adapt away from, or one dataset simply having more headroom for improvement than the other.

4.9.3 Replication on a Second Denoising Dataset (DNS → REVERB)

To confirm the finding does not depend on the specific choice of denoising dataset, the transfer was replicated using DNS English as the adaptation domain, evaluated on REVERB Challenge:

The pattern is essentially identical to the VoiceBank+DEMAND case (PESQ 2.50–2.70, SRMR 8.46–9.33), confirming the effect generalizes across denoising datasets rather than being specific to VoiceBank+DEMAND.

Table 4.17: Cross-task transfer replication: adapt on DNS (English), evaluate on REVERB Challenge.

Model	Method	PESQ↑	SISDR↑	FWSEG↑	LLR↓	CD↓	SRMR↑
Mixed v1	MPol	2.496	8.800	**10.209**	0.650	3.893	8.461
Mixed v2	MPol	**2.671**	**6.867**	**10.514**	**0.718**	**3.543**	**9.326**
Mixed v1	LaDen	2.510	8.590	9.391	**0.591**	3.918	**8.534**
Mixed v2	LaDen	2.683	5.867	9.865	0.736	3.577	9.130
Mixed v1	RemixIT	**2.519**	**8.852**	9.806	0.617	**3.826**	8.503
Mixed v2	RemixIT	2.701	6.669	9.879	0.781	3.628	8.855

4.10 Replication with Training-Distribution Denoising Data

A further replication used the models' own training-distribution denoising data (EARS-WHAM or EARS_DEMAND) as the adaptation domain, evaluated on synthetic reverb constructed by convolving EARS speech with ACE room impulse responses (EARS+ACE). Two variants of this experiment are reported below: **Variant B** adapts once, on the denoising data, and evaluates directly on EARS+ACE with no further adaptation, mirroring the cross-task procedure used throughout this section. **Variant A** takes Variant B's already-adapted checkpoint and applies a second, direct adaptation step on EARS+ACE itself before evaluating, testing whether adding the seemingly natural extra step of adapting on the actual target domain improves on Variant B's result.

Table 4.18: Source model on EARS+ACE, no TTA baseline.

Model	PESQ↑	SISDR↑	FWSEG↑	LLR↓	CD↓	SRMR↑
Isolated Reverb	1.681	-15.735	2.707	0.828	5.255	7.561
Mixed v1	2.501	8.675	10.469	0.536	3.783	8.327
Mixed v2	2.657	6.432	10.156	0.583	3.463	9.479

The Isolated Reverb model fails badly on this synthetic reverb condition (PESQ 1.68), while the mixed models, despite having no TTA applied here, already perform far better (PESQ 2.50–2.66), underscoring the greater robustness of jointly-trained models (this point is developed further in Chapter 5, H1).

(MPol shown above; LaDen and RemixIT show the same pattern, PESQ 2.42–2.69 across both datasets and both models.)

RIRmixIT was also tested in this same configuration, since it is the one method purpose-built for reverb and had otherwise underperformed source in every direct-reverb setting (Sections 4.5 and 4.7):

Cross-task transfer holds for RIRmixIT as well, with results comparable to the other three methods (PESQ 2.44–2.68). The cross-task effect is not specific to methods whose core assumption breaks under reverb (Section 5.3); even a method with a structurally correct reverb

Table 4.19: Variant B: adapt on EARSWHAM/EARS_DEMAND, evaluate on EARS+ACE, no second adaptation step (MPol shown).

Model	Dataset	PESQ↑	SISDR↑	FWSEG↑	LLR↓	CD↓	SRMR↑
Mixed v1	EARSWHAM	2.478	8.574	10.511	0.545	3.825	8.199
Mixed v2	EARSWHAM	2.623	6.523	10.324	0.555	3.497	9.323
Mixed v1	EARS_DEMAND	2.482	8.578	10.515	0.552	3.823	8.313
Mixed v2	EARS_DEMAND	2.638	6.634	10.385	0.584	3.511	9.367

Table 4.20: RIRmixIT, adapt on EARSWHAM/EARS_DEMAND, evaluate on EARS+ACE, no second adaptation step.

Model	Dataset	PESQ↑	SISDR↑	FWSEG↑	LLR↓	CD↓	SRMR↑
Mixed v1	EARSWHAM	2.470	8.940	10.226	0.623	3.939	8.326
Mixed v1	EARS_DEMAND	2.438	9.196	10.115	n/a	4.033	8.036
Mixed v2	EARSWHAM	2.671	6.002	9.889	0.651	3.494	9.148
Mixed v2	EARS_DEMAND	2.676	6.579	10.400	0.655	3.512	9.213

model shows the same pattern of large gains from indirect adaptation and no benefit from direct reverb adaptation. LLR is not available for the Mixed v1, EARS_DEMAND condition.

Cross-task transfer holds even when the adaptation dataset is the model's own training-distribution denoising data. EARSWHAM and EARS_DEMAND give near-identical results, further supporting that the effect is not sensitive to the specific denoising dataset used.

Table 4.21: Variant A: with a second adaptation step (MPol) on EARS+ACE, for comparison.

Model	Dataset	PESQ↑	SISDR↑	FWSEG↑	LLR↓	CD↓	SRMR↑
Mixed v1	EARSWHAM	2.040	5.416	7.788	0.787	4.554	7.288
Mixed v2	EARSWHAM	2.240	8.136	9.555	0.716	4.167	7.944
Mixed v1	EARS_DEMAND	2.046	5.407	7.877	0.787	4.522	7.331
Mixed v2	EARS_DEMAND	2.046	5.407	7.877	0.787	4.522	7.331

Adding a second, direct MPol adaptation step on EARS+ACE consistently makes results worse than Variant B (PESQ drops to 2.04–2.24). This is the first of three independent confirmations that a second direct-adaptation step on the target domain hurts rather than helps; the other two are the RemixIT+RIRmixIT chaining result (Section 4.11) and the sequential ordering result (Section 4.8). The implications are unified in Chapter 5.

4.11 RemixIT + RIRmixIT Chaining

As a further test of whether a second, reverb-appropriate adaptation step could improve on cross-task transfer alone, RemixIT was applied first (on VoiceBank+DEMAND), followed by RIRmixIT as a second step on REVERB Challenge, using RIRmixIT specifically because,

unlike MPol in the Variant A experiment above, its assumption is structurally correct for reverb.

Table 4.22: RemixIT + RIRmixIT chaining: RemixIT on VoiceBank+DEMAND, then RIRmixIT on REVERB Challenge.

Model	PESQ↑	SISDR↑	FWSEG↑	LLR↓	CD↓	SRMR↑
Mixed v1	1.905	-6.250	6.221	0.504	4.218	4.865
Mixed v2	1.691	-20.012	5.227	0.672	4.242	6.113

Even with a reverb-appropriate second step, chaining performs far worse than the cross-task transfer result alone (Section 4.9: PESQ 2.50–2.62 for Mixed v1/v2 with RemixIT). Mixed v2 shows a severe SI-SDR collapse to −20.0, indicating serious instability. This is the third independent confirmation, alongside Section 4.8 and the Experiment 4 Variant A result above, that adding any second direct-adaptation step on the target domain, even with a method whose assumptions are theoretically appropriate for that domain, degrades rather than improves on the cross-task result. This point is unified in Chapter 5.

A caveat on the generality of the cross-task transfer results in this section: all of the numbers reported above use CMGAN models trained as part of this project. A preliminary check was also run using a separately provided pretrained mixed-domain CMGAN checkpoint, trained by the supervisor rather than as part of this project, and showed a substantially weaker cross-task transfer effect than this project's own models. This discrepancy has not been explained and is discussed further as a limitation in Chapter 6; it is flagged here, rather than only at the end of the report, since it bears directly on how confidently the cross-task transfer effect established in this section can be claimed to generalize beyond the specific models this project trained.

4.12 LaDen Reverb Encoder Analysis

A reverb-specific WavLM mapping matrix (`WavLM_EARS_reverb_map.th`) was generated via the DIET pipeline on EARSRev clean/reverberant pairs, using the WavLM checkpoint provided by the supervisor. Baseline embedding similarity was 0.87, rising to an estimated 0.97 after mapping, confirmed correct by the supervisor and independently re-verified.

Note: a separate earlier run of the same cross-task direction with the reverb matrix reported higher PESQ for Mixed v1/v2, 2.4976/2.6318 versus the 2.2848/2.4820 shown above. This discrepancy between two runs of the same nominal configuration is unreconciled; the values in this table are reported as measured, and the H5 hypothesis discussed in Chapter 5 is held with correspondingly lower confidence as a result.

Contrary to expectation, the reverb-specific matrix performs *worse* than the original denoising matrix in direct reverb TTA. This paradoxical result (and its apparent reversal in the cross-task direction in at least one run) is examined in Chapter 5 (H5).

Table 4.23: Direct reverb TTA (REVERB Challenge): denoising matrix vs. reverb matrix.

Model	Matrix	PESQ↑	SISDR↑	FWSEG↑	LLR↓	CD↓	SRMR↑
Isolated Reverb	Denoising	**1.609**	**-6.842**	**5.085**	0.566	**4.806**	**5.082**
Isolated Reverb	Reverb	1.527	-7.556	4.400	**0.556**	4.900	4.979
Mixed v1	Denoising	**1.835**	**-6.571**	**6.286**	0.532	**4.261**	5.057
Mixed v1	Reverb	1.711	-7.215	5.274	**0.508**	4.489	**5.057**
Mixed v2	Denoising	**1.800**	-12.049	**5.979**	0.564	**4.128**	**6.128**
Mixed v2	Reverb	1.662	**-11.602**	4.904	**0.549**	4.419	6.106

Table 4.24: Cross-task transfer direction (adapt on denoising, evaluate on REVERB): denoising matrix vs. reverb matrix.

Model	Matrix	PESQ↑	STOI↑	CSIG↑	CBAK↑	COVL↑	SISDR↑
Mixed v1	Denoising	**2.482**	**0.927**	**3.817**	**2.899**	**3.136**	**7.927**
Mixed v1	Reverb	2.285	0.924	3.649	2.619	2.957	7.701
Mixed v2	Denoising	**2.603**	0.929	3.838	**2.634**	**3.179**	3.096
Mixed v2	Reverb	2.482	**0.931**	**3.844**	2.537	3.131	**5.901**

4.13 Isolated Model Investigation

The "no gains from denoising TTA on the isolated model" finding (Section 4.4) raised an obvious follow-up question: was this specific to the VoiceBank+DEMAND dataset, or a more general property of the isolated model? To find out, the isolated denoising model was additionally tested with MPol, LaDen, and RemixIT on DNS and EARS_DEMAND.

Table 4.25: Isolated denoising model, additional TTA evaluation on DNS and EARS_DEMAND.

Method	Dataset	PESQ↑	Source PESQ	Result
MPol	DNS	2.350	2.369	No gain
LaDen	DNS	2.309	2.369	No gain
RemixIT	DNS	2.369	2.369	No gain
MPol	EARS_DEMAND	2.478	~2.33	PESQ gain, but SSNR/SISDR collapse
LaDen	EARS_DEMAND	2.397	~2.33	Mixed result

DNS shows no gain from any method, consistent with the VoiceBank+DEMAND result. EARS_DEMAND shows a PESQ improvement for MPol and LaDen, but with a substantial drop in SSNR and SI-SDR, a trade-off rather than a clean improvement. This confirms the no-gains finding is a general property of the isolated denoising model, not an artifact of the VoiceBank+DEMAND setup. It also supports the H1 hypothesis developed in Chapter 5: the gap between isolated and jointly-trained models reflects something structural about joint training itself, not a quirk of one dataset.

5 Analysis and Hypotheses

5.1 Source Model Comparison: Do We Get Gains?

Before interpreting any hypothesis, it is worth establishing plainly which experiments in Chapter 4 actually beat the source model, and which do not.

Table 5.1: Summary: which experiments beat the source model.

Chapter 4 section	Experiment	Method	Source PESQ	Adapted PESQ	Beats source?
4.4	Denoising TTA, isolated model (VoiceBank)	RemixIT	2.62	2.62	No
4.4	Denoising TTA, isolated model (DNS)	MPol	2.37	2.35	No
4.13	Denoising TTA, isolated model (EARS_DEMAND)	MPol	~2.33	2.48	Mixed (metric trade-off)
4.5	Direct reverb TTA, isolated reverb model	RIRmixIT	1.89	1.71	No
4.7	Denoising TTA, mixed models (VoiceBank)	RemixIT	2.14	2.63	Yes
4.7	Direct reverb TTA, mixed models (REVERB)	MPol	1.89	1.84	No
4.9	Cross-task: adapt denoise → eval reverb	MPol	1.89	2.50–2.62	Yes
4.9	Cross-task: adapt reverb → eval denoise	MPol	2.14	2.66	Yes
4.10	Training-distribution denoise → EARS+ACE (no 2nd step)	RemixIT	~1.89	2.67–2.69	Yes

A clear pattern emerges from this table: this project only beats source under two conditions, adapting mixed models on denoising data, or doing cross-task transfer. Direct reverb adaptation never beats source, in any configuration tested, and isolated-model denoising TTA gives no clean gains either. The rest of this chapter is essentially an explanation of this table, one hypothesis at a time.

5.2 Why Denoising TTA Improves Reverb (H1)

H1: General representation improvement, enabled by joint training

The most striking result in this project (Section 4.9) is that adapting on denoising data and evaluating directly on reverb, with no reverb-specific adaptation at all, outperforms adapting directly on reverb (Section 4.5). This holds across all three TTA methods, both mixed models, and multiple denoising datasets (VoiceBank+DEMAND, DNS, EARSWHAM, EARS_DEMAND), and in both directions (denoise→reverb and reverb→denoise).

A closely related observation is why mixed models respond so much more strongly to denoising TTA than the isolated denoising model, given identical TTA methods and identical adaptation data (Section 4.4 vs. Section 4.7): the isolated model shows essentially no gain on VoiceBank+DEMAND (PESQ 2.62→2.62), while Mixed v2 shows a large gain on the same dataset (PESQ 2.14→2.63).

Hypothesis (unified): TTA does not simply suppress additive noise. It sharpens the model's internal representation of clean speech by updating normalization statistics in the TSCB con-

former blocks (see Figure 2.1 for where these blocks sit in the architecture). This representation is only *shared* across tasks, and thus only available for TTA to exploit in this way, when the model was jointly trained on both denoising and reverb (EARSCombined, 50/50). Joint training forces the TSCB blocks to develop task-agnostic clean-speech representations; isolated training specializes the model for one task only, leaving no shared structure for TTA to sharpen. This is why the isolated model shows no gain on the same denoising data that produces large gains in the mixed models.

The distinction being drawn here is analogous to the difference between learning a general skill and learning a task-specific trick. A model trained jointly on denoising and reverberation must find internal representations useful for recognizing clean speech in general, since that recognition problem is shared across both tasks it was trained on. A model trained on denoising alone has no such pressure and is free to specialize narrowly on whatever patterns happen to separate speech from additive noise in its own training distribution, without that specialization needing to generalize to anything else. TTA, on this hypothesis, does not teach the model something new; it sharpens whichever representation the model already has. If that representation is the general, task-agnostic one produced by joint training, sharpening it helps on both tasks. If it is the narrow, denoising-specific one produced by isolated training, there is nothing general there for TTA to sharpen, which is consistent with the isolated model showing no such transfer.

This hypothesis is supported by a weight delta analysis comparing TTA-adapted checkpoints against supervised finetuning checkpoints (5 epochs, VoiceBank+DEMAND, both mixed models). The analysis shows that TTA updates exclusively affect BatchNorm running statistics (`running_var`, `running_mean`) inside the Two-Stage Conformer Block (TSCB) layers, with zero change to any learned weight. These running statistics are not weights the model learns via backpropagation; they are running estimates of the mean and variance of the feature values passing through a given layer, tracked so the layer can standardize its inputs to a consistent scale. Updating only these statistics, without touching any learned weight, changes how the network normalizes and rescales the features flowing through it, but leaves every actual learned transformation (every convolution kernel, every attention weight, every decoder layer) exactly as it was after training. This is what makes the change "shallow": it recalibrates the network's internal operating point rather than teaching it anything new, which is consistent with a general, task-agnostic adjustment rather than a task-specific correction. Supervised finetuning, by contrast, changes actual learned weights (decoder biases, discriminator layers) in addition to the same normalization statistics at roughly twice the magnitude TTA achieves. This distinction (TTA shifts the internal feature *distribution*, finetuning shifts the learned *transformation*) is consistent with TTA producing a general, task-agnostic representation shift rather than a task-specific correction. Figure 5.1 below visualizes this per layer.

Additional support: only 6,020 of 2,016,483 parameters (0.3%) are updated by any of the four TTA methods, yet the cross-task effect is large, a small, shared normalization change producing a broad functional effect is consistent with the statistics capturing something fundamental about clean speech distribution rather than something noise-specific.

Further support comes from the EARS+ACE source comparison in Section 4.10: the Isolated Reverb model fails badly on this unseen reverb condition (PESQ 1.68) even without any TTA involved, while the mixed models handle it far better (PESQ 2.50–2.66) with no adaptation

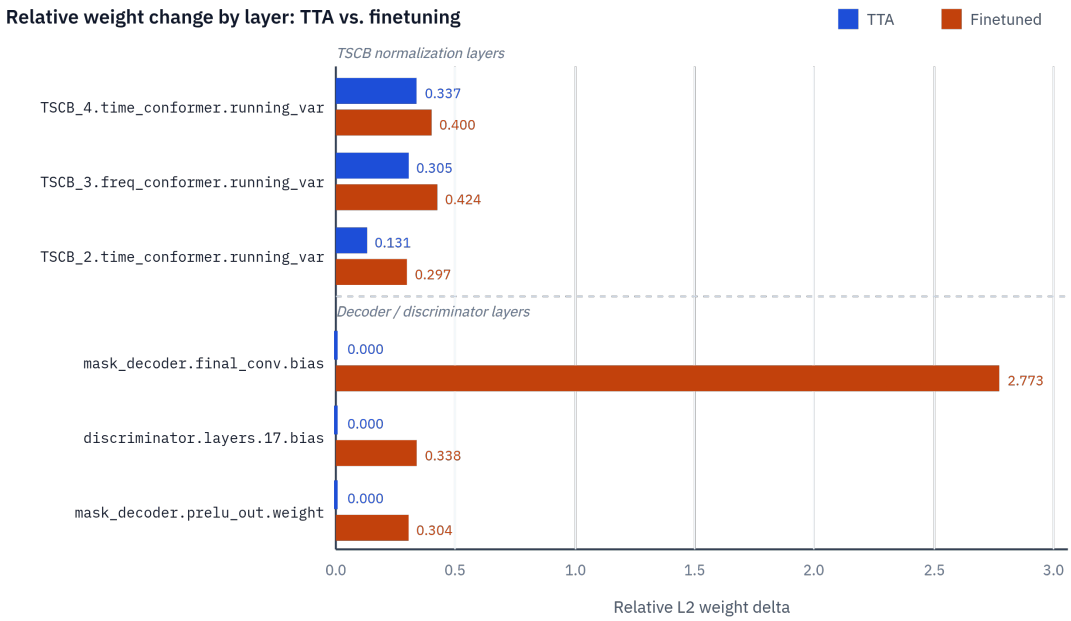


Figure 5.1: **Weight Delta by Layer.** TTA-adapted vs. finetuned checkpoints (Mixed v1, VoiceBank+DEMAND).

at all, evidence that joint training itself, independent of TTA, produces more robust representations.

5.3 Why Direct Reverb TTA Fails (H2)

Section 4.5 showed that all TTA methods, including RIRmixIT, designed specifically for reverb, hurt performance on the isolated reverb model relative to source. Section 4.7 showed this extends to mixed models as well.

A common thread runs through all three failures: each method carries an implicit assumption about the structure of the distortion it is correcting, and that assumption holds for additive noise but breaks down for convolutive reverb.

Diagnostic evidence for each method's failure mode was drawn from the adaptation loss curves during reverb TTA:

- **MPol:** the Sign metric (fraction of bins with confident bimodal mask predictions) collapses roughly 8-fold on reverb relative to denoising (0.00577 → 0.000717), directly showing the bimodal assumption is largely absent under reverb.
- **LaDen:** the Rate metric (fraction of samples accepted for adaptation) stays close to 0.998 on reverb. The model is confidently adapting on nearly every sample, but in the wrong direction, since performance still degrades. This indicates silent miscalibration rather than a visibly failing loss.
- **RemixIT:** the adaptation loss collapses roughly 500-fold on reverb relative to denoising (0.000387 → 7.73×10^{-7}), consistent with reverb data being trivially, and degen-

Table 5.2: TTA method assumptions and their applicability to reverb.

Method	Assumption	Holds for noise?	Holds for reverb?
MPol	Bimodal mask (speech bins vs. noise bins)	Yes	No, reverb smears energy across all bins
LaDen	Linear latent mapping, $A \cdot \text{WavLM(noisy)} \approx \text{WavLM(clean)}$	Approximately	No, convolution is not linear in embedding space
RemixIT	Additive, separable distortion	Yes	No, reverb is convolutive, not additive
RIRmixIT	Explicit RIR-based convolutive model	N/A	Correct model, but still underperforms source

erately, "separable" under RemixIT's additive assumption, producing a spuriously low loss that does not correspond to genuine improvement.

Even RIRmixIT, whose assumption is structurally correct for reverb, does not close the gap to source. This suggests that having the right assumption is necessary but not sufficient. The small parameter budget (0.3% of the model, restricted to normalization and output layers) may simply be too shallow to correct reverb distortion directly, even with the right inductive bias.

5.4 Why the Second Step Always Hurts, and Why Ordering Doesn't Matter (H3, H4)

H3: A second adaptation step on the target domain hurts

Three independent experiments in Chapter 4 confirm that adding a second, direct adaptation step on the reverb domain, after a first step on denoising, consistently makes results worse than stopping after the first step:

1. **Section 4.8 (sequential TTA):** the second step does not improve on the cross-task baseline established by the first step alone.
2. **Section 4.10 (Experiment 4, Variant A vs. B):** Variant B (no second step) achieves PESQ 2.48–2.62; Variant A (with a second MPol step on EARS+ACE) drops to PESQ 2.04–2.24.
3. **RemixIT+RIRmixIT chaining** (Section 4.11): adapt with RemixIT on VoiceBank+DEMAND, then RIRmixIT on REVERB Challenge as a second step, Mixed v1 PESQ drops to 1.91; Mixed v2 shows a severe SI-SDR collapse to −20.

These three confirmations come from methodologically independent experimental setups (a sequential ordering test, a comparison against a variant with no second step, and a chaining of two entirely different TTA methods), rather than three measurements of the same underlying

test. Agreement across independent setups is stronger evidence than the same finding repeated under near-identical conditions would be, since it rules out the result being an artifact specific to any one experimental design choice.

The likely explanation is that the second step's gradient updates conflict with the beneficial weight shifts achieved by the first step. The first step moves the model into a state with improved, more general representations (per H1, Section 5.2); the second step then applies a low-quality or actively harmful reverb-specific gradient (per the H2 diagnostics in Section 5.3), which partially undoes the first step's gains.

H4: Task ordering does not matter

Section 4.8 additionally shows that when both adaptation steps *are* performed, the order in which they are applied makes essentially no difference to the final result. Denoise-first and reverb-first converge to nearly identical PESQ for both tasks (Mixed v1: VoiceBank ~2.48–2.49 either way, REVERB ~1.83–1.84 either way).

A plausible explanation is that because all methods adapt only 0.3% of the model's parameters, restricted to normalization and output layers, the weight perturbation from each individual step is small enough that the two steps behave approximately additively and commutatively. The final state is dominated by the base model rather than by either adaptation step individually.

Reconciling H3 and H4

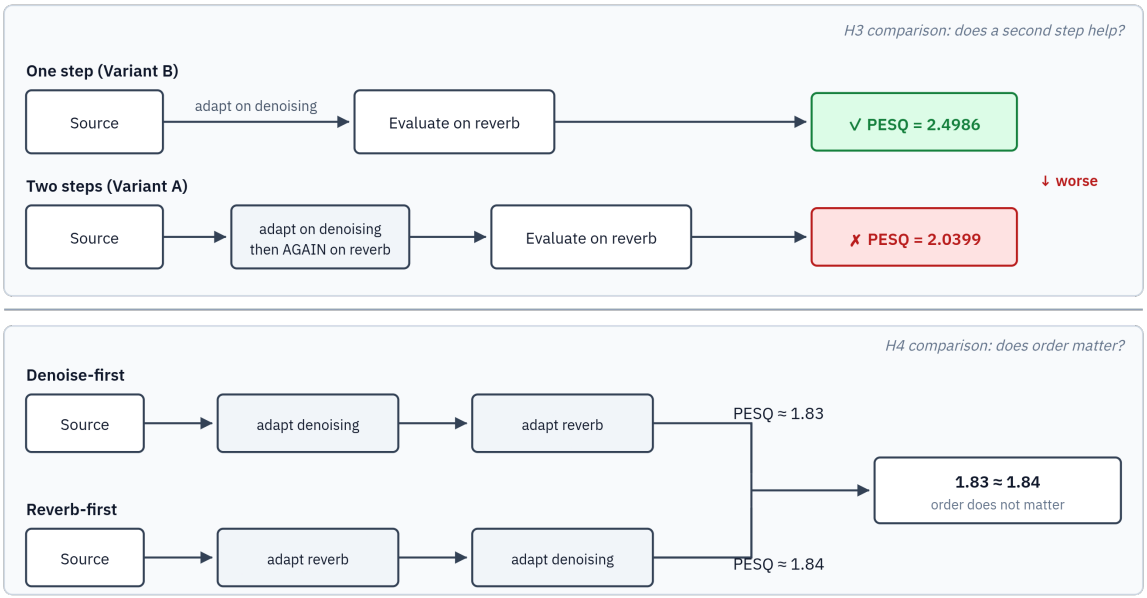


Figure 5.2: **H3 vs. H4 Reconciliation.** One step vs. two steps (H3), and two orderings of the same two steps (H4).

These two findings might initially appear to conflict: H3 says a second step is harmful, while H4 says the order of two steps doesn't matter. But they answer different questions, illustrated in Figure 5.2 above. H3 compares *one step vs. two steps* (two is consistently worse). H4 compares *ordering, given that two steps are always performed* (order does not matter). The unifying explanation is that reverb TTA reliably produces a low-quality, harmful gradient signal (Section 5.3) regardless of when it is applied in a sequence, so whichever order is used, the same harm is eventually absorbed, and the two orderings converge to a similarly degraded outcome.

5.5 The Reverb Matrix Paradox (H5)

Unlike H1 through H3/H4, which build on one another as a connected chain of reasoning about why cross-task transfer works and what its limits are, H5 addresses a separate, narrower question specific to one method (LaDen) and is held with correspondingly lower confidence, for reasons made explicit at the end of this section. It can be read independently of the preceding hypotheses.

Section 4.12 introduced a reverb-specific WavLM DIET matrix for LaDen, generated on EARSRev pairs and independently confirmed correct. Its behavior is context-dependent: in direct reverb TTA it performs *worse* than the original denoising matrix (isolated reverb PESQ 1.53 vs. 1.61), but in the cross-task direction it performs *better* in at least one measured run (Mixed v1 PESQ 2.50 vs. 2.35 with the denoising matrix). A separate run of the same configuration did not reproduce this reversal (see the note in Section 4.12), so this finding should be treated as provisional pending reconciliation.

Hypothesis: the denoising matrix maps a well-conditioned linear problem (speech-plus-additive-noise embeddings to clean-speech embeddings) with a stable inverse. The reverb matrix instead maps reverberant embeddings to anechoic embeddings, a convolutive relationship that is inherently harder to approximate with a single linear map. A mapping is described as poorly-conditioned when small differences in its input can produce disproportionately large differences in its output, making the mapping unstable to estimate and unstable to apply; a well-conditioned mapping does not have this sensitivity. Applied directly to a reverb-adaptation task from the source model's starting point, this poorly-conditioned mapping destabilizes the adaptation loss. Applied after the model has already been shifted into a better state by prior denoising adaptation, the same mapping may provide a useful, smaller additional correction that the model can accommodate.

Given the unresolved discrepancy between the two measured cross-task runs, this hypothesis is held with lower confidence than H1–H3, and the true value of the affected condition, 2.2848/2.4820 or 2.4976/2.6318 for Mixed v1/v2, remains undetermined.

5.6 Isolated Model Investigation

Section 4.13 took up the question of whether the isolated denoising model's lack of TTA gains (Section 4.4) was an artifact specific to the VoiceBank+DEMAND dataset, or something more general. Testing on DNS confirmed no gains for any method, matching the Voice-

Bank+DEMAND result. Testing on EARS_DEMAND showed a PESQ improvement for MPol and LaDen, but with a substantial trade-off in SSNR and SI-SDR.

This confirms the no-gains finding is a general property of the isolated denoising model, consistent with the H1 explanation above: without joint training on a second task, there is no shared representation for denoising TTA to sharpen beyond what the model already learned during training, and the model was already close to its ceiling on the shift datasets tested in Section 4.3.

6 Conclusion

6.1 Summary of Findings

This project set out to test whether test-time adaptation methods built for denoising domain shift carry over to dereverberation, and, more broadly, whether a model's response to TTA depends on being trained on a single distortion type or on both together. The results answer both questions, and the answer to the second turns out to explain the first.

The starting point is a negative one: none of the three existing denoising-oriented TTA methods tested, MPol, LaDen, and RemixIT, improve performance when pointed directly at reverberation. All three make it worse (Section 4.5). A reverb-oriented variant developed for this project, RIRmixIT, does better than the other three but still cannot match the untouched source model. The reason, developed as the H2 hypothesis in Section 5.3, is that each method's adaptation signal is built on an assumption, bimodal masking, a linear latent mapping, additive separability, that simply does not hold once the distortion is convolutive rather than additive.

A second finding explains why some of the adaptation attempts that do work, work as well as they do. Models trained jointly on denoising and dereverberation respond far more strongly to denoising TTA than models trained on denoising alone, even with identical methods and identical adaptation data (compare Section 4.7 with Section 4.4). A weight-level analysis (Figure 5.1) points to why: TTA's updates concentrate almost entirely in the normalization statistics of the model's shared conformer blocks, and those statistics are only useful for adaptation once the blocks have learned a task-agnostic representation, which appears to require joint training in the first place (H1, Section 5.2).

The central result of the project follows from putting these two findings together. Take a jointly-trained model, adapt it on one distortion type, and evaluate it directly on the other, with no adaptation step on that second type at all. This produces substantially larger gains than adapting the model directly on the distortion it is being evaluated on (Figure 4.1). The effect holds across all three TTA methods, in both directions of transfer, and across several denoising datasets (Section 4.9); it even holds for RIRmixIT, whose assumptions are, on paper, the right ones for reverberation. Adding a second, direct adaptation step on the target distortion after this transfer does not help matters and, confirmed three separate ways (H3, Section 5.4), reliably makes results worse. When two adaptation steps are performed regardless, the order in which they happen makes little difference to the outcome (H4, same section).

6.2 Limitations

The findings above should be read alongside a few limitations that are worth stating plainly rather than smoothing over.

The first concerns the reverb-specific LaDen matrix. Its behavior under cross-task transfer was measured in two separate runs of the same nominal configuration, and the two runs disagreed (Section 4.12). That discrepancy was never reconciled, which is why the corresponding hypothesis, H5, is held with lower confidence than the others. Two experiments planned to test H5 further, a direct comparison of LaDen's loss magnitude across matrix conditions and an ablation of its envelope regularization term, did not get finished in time to be included here.

The most consequential limitation concerns how far the H1 hypothesis can be trusted to generalize. All of the cross-task transfer results reported here used CMGAN checkpoints trained specifically for this project. A preliminary check with a separately provided pretrained mixed-domain checkpoint, trained by the supervisor on a training distribution not directly verified against this project's own, showed a noticeably weaker transfer effect. Why the two disagree is still an open question, and it cannot currently be ruled out that the central finding of this project depends on some particular property of how these models happened to be trained, rather than being a general property of jointly-trained CMGAN models under this adaptation procedure. This caveat was already flagged at the end of Section 4.9; it is repeated here because it matters enough not to be buried in a subsection.

Finally, all four TTA methods evaluated here were restricted to updating only normalization and output layers, following the design of the published methods this project builds on. Whether the ordering-independence finding (H4) would still hold under a less restricted adaptation budget was not tested.

6.3 Future Work

Several of the limitations above point directly to what should be done next.

The most obvious gap is a TTA method actually designed around convolutive distortion rather than adapted from an additive one. RIRmixIT was a first attempt in this direction, but it still falls short of the untouched source model even though its assumptions are structurally the right ones for reverberation. That a reverb-appropriate method still underperforms suggests the shallow adaptation budget shared by every method tested here, and not just the choice of assumption, may itself be holding results back. A systematic comparison of adaptation budgets, from the norm-and-output-only restriction used throughout this project up to full-model adaptation, would help settle whether the ordering-independence finding (H4) is a genuine property of TTA under domain shift or simply an artifact of how narrow the parameter budget was.

Beyond adaptation design, two open questions are more about scope than method. Testing on reverberation datasets beyond REVERB Challenge and the synthetic EARS+ACE condition would clarify how much of what was found here is specific to the reverberant conditions actually tested. And resolving the unexplained gap in cross-task transfer strength between this

project's own trained models and the separately provided pretrained checkpoint remains the single most important open question: it would clarify whether the joint-training mechanism behind H1 depends on some particular detail of how these models were trained, or whether joint training on both tasks is enough on its own.

Appendix: Use of AI Tools

The author of this project used an AI tool, specifically Claude (Anthropic). It was used in three ways. First, and most substantially, for engineering support: the tool helped build comprehensive configuration files for the training and test-time adaptation experiments, and helped optimize the batch scripts used to run several TTA experiments in parallel on the cluster. Second, for progress tracking, once each week's experimental work had been completed by the author: the tool was used to compile a summary of that work into a running memory record, which the author used to check what had and had not yet been done. Third, on the manuscript itself, after the text had been written by the author: the tool was used through an agentic coding interface to edit the LaTeX sources and lay the document out quickly, and in the same pass to paraphrase and copy-edit drafts.

All experimental design, execution of training and test-time adaptation experiments on the cluster, generation of the underlying experimental results, and the analysis, hypotheses, and conclusions drawn from them were carried out solely by the author, who has verified all quantitative claims against the underlying experimental output and assumes full responsibility for the entire content of this work.

Code Availability. The full codebase for this project, including the evaluation pipeline and all TTA method implementations, is available at https://github.com/rustombhesania/SpeechEnhancement.

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